

exactly 80 mm: \_\_\_\_\_

## FIRST SOLVE

New here? Do the walkthrough once at [cubepath.com/vars.vercel.app/](#)  
 Your solve count is the memory jog.

### NOTATION

R U F L B = turn that face clockwise, looking at from outside the cube  
 r = counter-clockwise, z = half turn, Lowercase r f = that face plus the layer behind it, turning together.  
 y = spin the whole cube left, white staying down. A whole-cube move isn't notation.

### WORDS

algorithm = a fixed sequence of turns - trigger = a short algorithm your hands run as one unit

## 1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm – work it out. Can't see it? Build the four white edges on the YELLOW face first, each above its matching center; then turn each one down 180 degrees.

 **R** UR'U' sexy move

 **G** UR'U' Sune

 **B** FRF' sledgehammer

gap = change grip

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## 2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the slot it belongs in, then match a picture below and repeat until it drops in.



**white sticker RIGHT**  
**RU'R'**



**white sticker FRONT**  
**L'L'U**



**white sticker UP** run the first one once to knock it out, then look again  
**RU'R'**



**3 MIDDLE EDGES**  
Top edge with NO yellow: Turn U until its front color matches the center below it. The arrow shows which slot it goes to.

goes **RIGHT**  
**U RU'U' y L'U'L'U**

goes **LEFT**  
**U' L'U'L'U y' R'U'R'**

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

### FIRST SOLVE

New here? Do the walkthrough once at [cubepath-six.vercel.app/](http://cubepath-six.vercel.app/) learn. This course is the memory jog.

### NOTATION

**R** = turn that face clockwise, looking at from outside the cube  
**C** = counter-clockwise, **2** = half turn. Lowercase **r** = that face plus the layer behind it, turning together.  
**y** = spin the whole cube left, white staying down. A whole-cube move is not fixing.

### WORDS

algorithm = a fixed sequence of turns - trigger = a short algorithm your hands run as one unit

## 1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's color must match the center it touches. No algorithm - work it out. Can't see it? Build the four white edges on the YELLOW face first, each above its matching center, then turn each one down 180 degrees.

## 2 WHITE CORNERS

Find a white corner on top. Turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



**white sticker RIGHT**  
**R'U'R**



**white sticker FRONT**  
**L'U'L**



**white sticker UP** run the first one once to knock it out, then look again  
**R'U'R**

## 3 MIDDLE EDGES

Top edge with NO yellow: turn U until its front color matches the center under it. The arrow shows which slot it goes to.



**goes RIGHT**  
**U R'U'U' y L'U'L**



**goes LEFT**  
**U' L'U'L U' y' R'U'R**

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

■ **R**U'R sexy move   
 ■ **R**U'R Ume   
 ■ **R'FR'** sledhammer   
 gap = change grip

### TWO-LOOK

#### OLL CROSS - yellow edges



Line bar left-to-right  
**fRUR'U'f**

Hook hook pointing front-right  
**fRUR'U'f**

#### OLL CORNERS - yellow face



Sune  $\frac{1}{2}$  yellow, rest clockwise  
**RUR'U RU2R'**

Anti-Sune  $\frac{1}{2}$  yellow, rest counter-clockwise  
**RU2R' U'RUR'**

PI  $\frac{1}{2}$  yellow; headlights left  
**fRUR'U'f fRUR'U'f**

#### OLL CORNERS - continued



Headlights  $\frac{1}{2}$  yellow; headlights at the back  
**R2DR'U2 RD'R'U2R'**

2x Headlights  $\frac{1}{2}$  yellow; headlights both sides  
**RUR'U RU'R' URU2R'**

Chameleon  $\frac{1}{2}$  yellow next to each other  
**rUR'U r' FRF**

Bowie  $\frac{1}{2}$  yellow diagonal  
**fR'UR'U r' FR**

#### FAILBACK

The badge on each row is your failback: that many Sunes, with a U turn around, also finishes the case. Machine-checked, there is the worst.  
Top not all yellow and nothing matches? Run Sune and read again.

No yellow edge at all? Run the first one, then read again.

■ **RUR'U** sexy move   ■ **RUR'U** Sune   ■ **R'FR** sledgehammer   gap = change grip

### TWO-LOOK

**OLL CROSS** - yellow edges



Line bar left-to-right  
**fRUR'U'f**

Hook hook pointing front-right  
**fRUR'U'f**

**OLL CORNERS** - yellow face



Sune  $\frac{1}{2}$  yellow, rest clockwise  
**RUR'U RU2R'**

Anti-Sune  $\frac{1}{2}$  yellow, rest counter-clockwise  
**RU2R' U'RUR'**

P1  $\frac{1}{2}$  yellow; headlights left  
**fRUR'U'f fRUR'U'f**

### OLL CORNERS - continued



Headlights  $\frac{1}{2}$  yellow; headlights at the back  
**R2DR'U2 RD'R'U2R'**

2x Headlights  $\frac{1}{2}$  yellow; headlights both sides  
**RUR'U RUR'R URU2R'**

Chameleon  $\frac{1}{2}$  yellow next to each other  
**r'RUR'r' FRF**

Bowtie  $\frac{1}{2}$  yellow diagonal  
**f'RUR'U'r' FR**

**FAILBACK**

The badge on each row is your fallback: that many Sunes, with a U turn around, also finishes the case. Machine-checked; there is the worst.  
Top not all yellow and nothing matches? Run Sune and read again.

No yellow edge at all? Run the first one, then read again.

■ RUR'U sexy move    ■ RUR'U Sune    ■ R'FRF sledgehammer    ■ swap - change grip

### ONE-LOOK PLL

#### ADJACENT CORNER SWAP exactly one face shows headlight



**\*T.L. headlight; pairs B-left F-left**  
**RUR'U'R' R'F R2UR'U'R' RUR'F'**

**Ja L solid; pairs B-right F-left R-front**  
**x R2FRF' RU2 r'Ur U2**

**Jb L solid; pairs B-left F-right B-back**  
**RUR'F' RUR'U'R' R'F R2UR'U'**

**Aa L solid; pairs F-right R-front**  
**x L2D2 L'U'L D2 L'U'L**

**Ab L headlight; pairs B-right R-back**  
**x L2D2 LUL' D2 LU'L**

**F.L solid**  
**RUR'F' RUR'U'R' R'F R2UR'U'R'**

### ADJACENT CORNER SWAP continued



**Ra L. headlight; pairs B-left**  
**RUR'U'R' RUR DR'URD' R'U2R'**

**Rb L. headlight; pairs B-left**  
**R2F RUR'U'R' F'R U2R'U2R**

**Ga L. headlight; pairs F-right**  
**R2UR'UR'U' RU'R2 U'D R'URD'**

**Gb L. headlight; pairs B-back**  
**RUR' UD' R2UR'UR'U' RU'R2D**

**Ge L. headlight; pairs R-right**  
**RUR'U'RU RU'R2 UD' RU'RD**

**Gd L. headlight; pairs R-front**  
**RUR' UD' R2UR'UR'U' UR'UR2D**

One face shows headlight; two corners of that face match. Hold as shown, turn, then turn the top to finish.

**RUR'U' sexy, moves, B.RUR'U' sync, F.R'RF' sldochameleon, ego = chancey grin**

### ONE-LOOK PLL ADJACENT CORNER SWAP

exactly one face shows headlight



• T1: headlight; pairs B-Left F-Left  
**R'U'R' R'F R2U'R' RUR'**



Ja L solid; pairs B-Right F-Left R-front  
**x R2FRF' RU2 r'Ur U2**



Jb L solid; pairs B-Left F-right R-back  
**RUR' RUR' R'F R2UR'**



Aa L headlight; pairs F-right R-front  
**x L2D2 L'U' D2 L'U'L**



Ab L headlight; pairs B-Right R-back  
**x' L2D2 LUL' D2 LU'L**



F L solid  
**R'UF' RUR' R'F R2UR'UR'**

### ADJACENT CORNER SWAP

continued



Ra L1: headlight; pairs B-Left  
**RUR'UR' RUR DR'UR'D' R'U2R'**



Rb L1: headlight; pairs B-Left  
**R2F RURU' F'R U2R'U2R**



Ga L1: headlight; pairs F-right  
**R2UR'UR'U' RU'R2 U'D R'UR'D'**



Gb L1: headlight; pairs B-back  
**R'UR' UD' R2UR'UR' RU'R2D**



Ge L1: headlight; pairs B-right  
**R2UR'RU' RU'R2 UD' RU'R'D**



Gd L1: headlight; pairs R-front  
**RUR' UD' R2UR'RU' UR'UR2D'**

One face shows headlight; two corners of that face match. Hold as drawn on, then turn the top to finish.

Learn in the printed order, three a week, about six weeks. Ja and Jb on the same day – this one is the mirror of the other.

■ R'UR' sev.moves ■ R'UR' sev.moves ■ R'FRF' skidochance spin
■ R'UR' sev.moves ■ R'UR' sev.moves ■ R'FRF' skidochance spin

# ANALYSIS - BIG CUBES and THE MAP

**BIG CUBES** 4x4 and 5x5  
 Reduce: build the centers → pair the 12 edges → solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd Layer ONLY → never widen it.  
 last 2 edges  
 4x4 PLL parity  
 4x4OLL parity = 2 edge pair flipped - half the time  
 Rw U2 x Rw U2 Rw U2 Rw U2  
 Rw U2 x Rw U2 Rw U2 3Rw U2 Lw U2 Rw U2 Rw U2 Rw U2 Rw U2  
 Fix parity during the last layer, before OLL and PLL - both parity algorithms move corners.  
**NOTATION**  
 Each letter turns that face clockwise, seen from your outside the cube → so from your seat, D and B look counter-clockwise. = counter-clockwise. 2 = half turn.  
 M = middle slice, turning the way L turns. x y z = the whole cube, turning the way U and F turn. Lowercase f l = that face plus the layer behind it, turning together.  
**WORDS**  
 algorithm = a fixed sequence of turns - trigger = a short algorithm your hands run on as one unit - sexy move = the R U R' U' trigger -  
 sledgehammer = the R' F R' trigger - parity = a case a 3x3 cannot have - edge pair = two edges glued into one, on a 4x4 or 5x5  
 R U R' sexy move    R U R U S    R' F R' sledgehammer    gap = change grips

**NNEX - BIG CUBES AND THE MAP**

**BIG CUBES**    4x4 and 5x5  
 Reduce: build the 6 centers → pair the 12 edges → solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd Layer ONLY – never widen it.

last 2 edges  
 4x4 PLL parity     $Rw' \ U' \ R' \ U \ R' \ F \ R' \ F' \ Rw$  slice out, run it, slice back – both cubes  
                              $U2 \ L2 \ U2 \ U2 \ W2 \ U2 \ L2 \ U2 \ W2$  2 edge pairs swapped - half the time

4x4OLL parity – 1 edge pair flipped - half the time  
 $Rw \ U2 \ x \ Rw \ U2 \ Rw \ U2 \ Rw'$      $U2 \ L2 \ U2 \ Rw' \ U2 \ Rw \ U2 \ Rw' \ U2 \ Rw'$   
 Fix parity by turning the last layer, before OLL and PLL – both parity algorithms move corners.  
 one edge difference 3x3 PLL parity

**NOTATION**  
 Each letter turns that face clockwise, seen from outside the cube – so from your point A, D and B look counter-clockwise. = counter-clockwise. 2 = half turn.  
 M = middle slice, turning the way L turns. x y z = the whole cube, turning the way U, D and F turn. Lowercase f l = that face plus the layer behind it, turning together.

**WORDS**  
 algorithm = a fixed sequence of turns / trigger = a short algorithm you hands runs as one unit / sexy move = the R U R' U' trigger  
 sledgehammer = the R' F R' trigger / setup = a case a 3x3 cannot have / edge pair = two edges glued into one, on a 4x4 or 5x5

**TRURU** sexy move    **RURU**Sexy    **FRFP** sledgehammer    gap = change grip

Print at 100% / Actual size — do NOT fit to page. Two-sided, flip on the LONG edge. The number exactly 80 mm:

exactly 80 mm: \_\_\_\_\_

### 4 YELLOW CROSS

yellow edges only – ignore the corners



**Dot** no yellow edge at all  
**FRUR'U'F'**



**Hook** two at a right angle: hold them BACK and LEFT  
**FRUR'U'**



**Line** hold the bar left-to-right  
**FRUR'U'**

One algorithm, up to three times: dot to hook to line to cross.

### 5 MATCH THE YELLOW EDGES

First turn it so at least two top edges match the center under this. This algorithm is called *Sune*; you use it on every card after this one.



**two matched, side by side** hold them BACK and LEFT  
**FRUR'U' R2U'**



**two matched, opposite** run it once from any hold, then look again  
**FRUR'U' R2U'**

### 6 PUT THE YELLOW CORNERS HOME



**one corner home** hold it FRONT-LEFT  
**RU'L U' R'L**



**no corner home** run it once from any hold, then look again  
**RU'L U' R'L**

This cycles the other three corners into place. It also twists the yellow face you just look expected, the next step rebuilds it.

### 7 TWIST THE YELLOW CORNERS



**yellow faces RIGHT**  
**R'D'R D' R2**



**yellow faces FRONT**  
**D'R'DR x2**

Between corners run **ONLY** the top face, to bring the next unsolved corner to front-right. Never turn the whole cube. The cube will look distorted while you do this. Keep going – the last top turn brings it back.

**LAST-LAYER ORDER ON THIS CARD:** yellow cross, match edges, place corners, twist corners. Card 2 replaces this order permanently – it is the only card on this card that gets reset.

**UNDOCK CARD 2** – five scramble in a row with this card face down. **0 MASTER:** those five under 300. My target: \_\_\_\_\_ NEXT: 2/3 TWO-LOCK

### 4 YELLOW CROSS

yellow edges only – ignore the corners



**Dot** no yellow edge at all  
**FRUR'U'**



**Hook** two at a right angle: hold them BACK and LEFT  
**FRUR'U'**



**Line** hold the bar left-to-right  
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One algorithm, up to three times: dot to hook to line to cross.

### 5 MATCH THE YELLOW EDGES

First turn U so at least two top edges match the center under them. This algorithm is called **Sune**; you use it on every card after this one.



**two matched, side by side** hold them BACK and LEFT  
**FRUR'U' R2U'**

look again



**two matched, opposite** run it once from any hold, then  
**FRUR'U' R2U'**

look again

### 6 PUT THE YELLOW CORNERS HOME



**one corner home** hold a FRONT-LEFT  
**RU'L'U' R'L'**



**no corner home** run it once from any hold, then look again  
**RU'L'U' R'L'**

This cycles the other three corners into place. It also twists the yellow face you just moved. Expected, the next step rebuilds it.

### 7 TWIST THE YELLOW CORNERS



**yellow faces RIGHT**  
**R'D'R'D x2**



**yellow faces FRONT**  
**R'D'R'D x2**

Between corners turn **ONLY** the top face, to bring the next unsolved corner to front-right. Never turn the whole cube. The cube will look distorted while you do this. Keep going – the last top turn brings it back.

**LAST-LAYER ORDER ON THIS CARD:** yellow cross, match edges, place corners, twist corners. Card 2 replaces this order permanently – it is the only one this card that gets retired.

**UNLOCK CARD 2** – five scrambles in a row with this card face down. 3 MASTERS: those five under 300. My target: \_\_\_\_\_ NEXT: 2/3 TWO-LOCK

**PLL CORNERS** = headlight = 2 matching corners on one face



T headlights on ONE face = hold that face LEFT  
**RUR'U' R' R2UR'U' RUR'**  
**T-Perm**: No headlights to hold left? Run it twice.  
 Y no headlights = the two swapping corners sit diagonally  
**FRU'R' RUR' RUR' R'FR'**

**PLL EDGES** = corners home, top still not solved



UB the front edge travels LEFT  
**R2U RUR'U' R' UR'**



UB the front edge travels RIGHT  
**M2U M2U' M2UM2**



H both pairs swap straight across  
**M2UM2 U2 M2UM2**



Z each pair swaps with its neighbour  
**M'U' M2UM2M'U' M'U2 M2U**

**START HERE** = two new algorithms finish any last layer  
 Yellow cross: F R U R' U' until the cross appears -- at most 3.  
 Yellow face: Same, read again, repeat -- at most 3 (see the badges).  
 Corners: T-Perm. No headlights to hold left? Run it twice.  
 Edges: UB. No solved edge to put at the back? Run it twice.  
 Each case on the front replaces one repeat with one run. Three a week you can solve the whole four faces.

**HVSZ**  
 Both show headlights on all four faces. The edge between them decides: H belongs to the opposite face = H, to a neighbour = Z.

**STUCK?**  
 Corners home but nothing solved? Turn the top face once and read again -- that is often the whole lot.  
 One case has beaten you five times? Park it, use its fallback above, come back next week.  
 A single piece locks twisted and no algorithm touches it? The cube was reassembled wrong. Pop it out and restart it.

**WORDS**  
 OLL = make the whole top face yellow - PLL = slide the top pieces home - headlight = 2 matching corners on one face - ALUF = the free top turn that lives a case up

**PLL CORNERS:** headlights = two matching corners on one face



T headlights on ONE face – hold that face LEFT  
**RUR'U' R'F R2UR'U' RUR'F'**

Y no headlights – the two swapping corners sit naturally  
**FRU'R' RUR'F' RUR'U' R'F'R'**

**PLL EDGES:** corners home, top still not solved



Uh the front edge travels LEFT  
**R2U RUR'U' R'F' RUR'U'**

Ua the front edge travels RIGHT  
**M2U MU2 MU2M**

H both pairs swap straight across  
**M2UM2 U2 M2UM2**



Z each pair swaps with its neighbour  
**M'U' M2UM2U' M'U'U2 M2U**

**START HERE:** Two new algorithms finish any last layer.  
 Yellow cross: F R U R' U' until the cross appears – at most 3.  
 Yellow faces: Same, read again, repeat – at most 3 (see the badges).  
 Corners: F-Perm. No headlights to hold left! Run it twice.  
 Edges: U. No solved edge to put at the back! Run it twice.  
 Each case on the front replaces one four faces on one run. Three a week, you can solve the whole time.

**HVS Z:**  
 Both show headlights on all four faces. The edge between them decides if it belongs to the opposite face = H, to a neighbour = Z.  
**STUFF:**  
 Corners home but nothing solves? Turn the top face once and read again – that's often it.  
 One case has beaten you five times? Park it, use its fallback above, come back next week!  
 A single piece looks twisted and no algorithm touches it? The cube was reassembled wrong. Pop it out and reset it.

**WORDS**  
 OLL = make the whole top face yellow - PLL = slide the top pieces home - headlights = two matching corners on one face - AUF = the free top turn that lines a case up

[illegible]

### EDGES ONLY - corners already home

- **Ua B solid; F L R swap**; 3 edges counter-clockwise  
**M2U M2U M'U M2**
- **Ub B solid; F L R headlights**; 3 edges clockwise  
**R2U R'U R'U R'U R'U R'**
- **H all 4 headlights**; opposite edges swap  
**M2U'M2 U2 M2U'M2**
- **Z all 4 headlights**; adjacent edges swap  
**M'U M2U'M2U' M2U M2U**

### DIAGONAL CORNER SWAP - no headlights anywhere

- **Y pairs F-left R-back**  
**FRU'R'U' R'U'R'F' R'U'R'U' R'F'R'F'**
- **V pairs F-left L-front**  
**R'U'R'U' y R'F' R'ZU'R'U' R'F'RF'**
- **Na pairs R-right F-right L-back R-back**  
**R'U'R'U' R'U'R'F' R'U'R'U' R'F' R'ZU'R'U' U2R'U'R'**
- **Nb pairs B-right R-right L-back B-front**  
**R'U' URU'R' F'U'F' R'U'R' F'R'F' R'U'R'**
- **E nothing matches**  
**X' L'U'L'D' L'U'L'D' L'U'L'D' L'U'L'D'**

### HOW TO LOOK

- Count the **headlight** faces. One = front of this card. Four = this column. None = the column on the right.
- Only then read the edges to pick the exact case.
- Line the case up as drawn, run it, turn the top again to finish – that last free turn is the AUF.

• already yours from Card 2.

### WORDS

**PLL** = slide the top pieces home - headlights = 2 matching corners on one side; AUF = one free top turn that makes a case up-adjacent corner swap = the swapping corners share an edge - diagonal corner swap = the swapping corners sit opposite

Card 2 gives you 19/22 last layers in one look. This card gives 71/72. The other 51 off 72+ face cases, and the first two layers solved on pairs – drilled with a cube in the app: cubeapp+st3.vrered.app practices under 0.30. My target: all – twenty-two named correctly, twenty-one in a row, no pairs in hand, on two separate days; in 10 seconds.

## THE LADDER

1/3 FIRST LOOK - 5 algorithms - unlock: five solves, card face  
 2/3 TWO-LOOK - 11+ unlock: fifteen last letters in exactly two  
 3/3 ONE-LOOK PL - +15: end: all 21 named, twice on separate  
 days

Notes: that: full OLL, F2L, cross planning, look-ahead – in the app  
 not on card stock

## WHY THERE ARE ONLY THREE

1. The first two are the only ones you tell apart by sight. It is  
 bad at this you drill F2L, cross planning and look-ahead are drills  
 with no finite case tell. Full OLL is 57 cases, 22 of which differ only  
 by a diver 4 fifth of a millimetre wide at this size. All of them are  
 in the app, with a real cube and randomised setup. This set stops  
 where the medium stops

## PRINT AND VERSION

EVERY ALGORITHM on these cards is expanded from machine-verified data and checked against a cube simulator at build time; if one were  
 to wonder the build would follow the same path. Recognition wording is generated from the same permutation diagram is drawn from.

Set v1.0: **reprint** any single card at cubepalsh-us.vercel.app/print - print at 100% / Actual size, never Fit to page!

Not a tier, no unlock, no record. Use it when you buy a 4.

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THE LADDER
13 FIRST Solve - 9 algorithms - unlock: five solves, card face
down - 100% - Actual size, never Fit to page.
20 TWO-LOCK - 135 - unlock: fifteen last layers in exactly two
moves - 100% - Actual size, never Fit to page.
35 THREE-LOCK PLUS - 115 - done: all 21 named, twice on separate
days - 100% - Actual size, never Fit to page.
40 FOUR-LOCK PLUS - 115 - done: all 21 named, twice on separate
days - 100% - Actual size, never Fit to page.
WHY THERE ARE ONLY THREE
There are 128 possible orders of cases you tell apart by sight. It is
bad at this you drill. ZLL, cross planning and look-ahead are drills
with no finite case tall. Full OLL is 57 cases, 22 of which differ only
in a driver a fifth of a millimetre wide at this size. The 22 cases
are in the app, with a real cube and randomised setup. This sets stops
where the medium size is.
PRINT AND VERSION
EVERY ALGORITHM on these cards is expanded from machine-verified data and checked against a cube simulator at build time; if one were
wondering, you could build would find the card. Recognition wording is generated from the same pronunciation diagram is drawn from.
Set v1.0 - reprint any single card at cubepuh-35-vercelapp/print - print at 100% / Actual size, never Fit to page.
Not a tier, no unlock, no order. Use it when you buy a 4x4.

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Print at 100% / Actual size – do NOT fit to page. Two-sided, flip on the LONG edge. The number  
exactly 80 mm: \_\_\_\_\_